

Novelty Statement

Our paper addresses a fundamental question in biology: how do new ecologies and morphologies develop? The development of adaptive landscape thinking provided a way to cluster species into adaptive zones, but did not provide a pathway for how new regions of the landscape become populated. Our investigation suggests that transitions to new morphological and ecological spaces are facilitated by ecologically generalist and morphologically intermediate ancestors. Using new morphological and molecular datasets, we provide evidence of this pathway in Australian dragon lizards. This generalist-to-specialist transition provides a basis for how ecological opportunity can result in adaptive radiation, a central tenet of biodiversity research.



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30 March 2026
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Dear Prof. Akira S Mori,

We would like to begin by thanking both you and the two anonymous reviewers for your creative and valuable thoughts. On behalf of myself and my coauthors, we appreciate your time and contributions.

Reviewer 1 was fully satisfied with our previous responses and complimented the new analyses included. Reviewer 2 was also highly supportive of our updated manuscript, and provided a clever solution for how we could resolve their major outstanding critique. Here, we have incorporated their advice, principally fitting generalized linear models to link the ideas of habitat usage and morphology more clearly. We have also incorporated your suggestions regarding adjustments to figures that we think improve their value.

Below please find **reviewer comments in green**, and our responses in black. New text added to the manuscript is **highlighted**. We have formatted a page break between each reviewer. Where necessary, we have interjected our responses among reviewer comments.

On behalf of myself and coauthors, we thank you for your time and consideration,

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Generalists link peaks in the shifting adaptive
landscape of Australia's dragon lizards

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Conceptualization, IGB, JSK, NC; sampling, LGT, JS; data collection ESB, JT, IGB; data
curation, IGB, ESB; methodology, NC, IGB; analysis IGB, ST; writing - original draft, IGB;
writing - reviewing & editing, all authors; funding acquisition, JSK.

Data Accessibility: All data and code are available on GitHub:

www.github.com/IanGBrennan/Amphibolurinae

Running Title: Generalists link adaptive peaks

Keywords: adaptive landscape, macroecology, macroevolution, phylogenomics

Article Type: Letter

Word Counts: abstract (139), main text (4948)

Number of: references (67), figures (5), tables (0), text boxes (0)